

Protocol Layering

Application Layer

Function

- ▶ Provides network services directly to user applications.

Examples:

- ▶ Web browsing
- ▶ Email
- ▶ File transfer

Application Layer

Responsibilities

- ▶ Allows communication between application programs
- ▶ Sends user requests and receives responses
- ▶ Provides network services to applications

Application Layer

Example Protocols

- ▶ HTTP → Web browsing
- ▶ SMTP → E-mail service
- ▶ FTP → File transfer
- ▶ TELNET / SSH → Remote login
- ▶ DNS → Domain name resolution
- ▶ SNMP → Network management

Application Layer

Data Unit

► *Message*

Transport Layer

Function

- ▶ Provides process-to-process communication between hosts.

Transport Layer

Responsibilities

- ▶ Segmentation and reassembly
- ▶ Flow control
- ▶ Error control
- ▶ Congestion control
- ▶ Reliable or unreliable data transfer

Transport Layer

Main Protocols

TCP (Transmission Control Protocol)

- ▶ Connection-oriented
- ▶ Reliable communication
- ▶ Provides flow, error, and congestion control

UDP (User Datagram Protocol)

- ▶ Connectionless
- ▶ Faster with low overhead
- ▶ No flow or error control

SCTP (Stream Control Transmission Protocol)

- ▶ Designed for multimedia applications

Transport Layer

Data Unit

- ▶ *Segment / User Datagram*

Network Layer

Function

- ▶ Provides host-to-host communication and routing of packets across networks

Network Layer

Responsibilities

- ▶ Logical addressing
- ▶ Routing
- ▶ Packet forwarding
- ▶ Selecting the best path

Main Protocol

- ▶ IP (Internet Protocol)

Network Layer

Supporting Protocols

- ▶ ICMP → Error reporting
- ▶ IGMP → Group management
- ▶ DHCP → IP address assignment
- ▶ ARP → Address resolution

Features of IP

- ▶ Connectionless protocol
- ▶ No flow control
- ▶ No error control
- ▶ No congestion control

Network Layer

Devices Using This Layer

- ▶ *Routers*

Data Unit

- ▶ *Datagram*

Data Link Layer

Function

- ▶ Provides communication between devices on the same link.

Data Link Layer

Responsibilities

- ▶ Framing
- ▶ Link-level addressing
- ▶ Error detection/correction
- ▶ Node-to-node delivery

Data Link Layer

Features

- ▶ Encapsulates datagrams into frames
- ▶ Supports wired and wireless links

Devices Using This Layer

- ▶ *Switches*

Data Unit

- ▶ *Frame*

Physical Layer

Function

- ▶ Transmits bits as electrical, optical, or wireless signals through the transmission medium.

Physical Layer

Responsibilities

- ▶ Bit transmission
- ▶ Signal conversion
- ▶ Physical connection handling

Physical Layer

Transmission Media

- ▶ Copper cables
- ▶ Fiber optic cables
- ▶ Air (wireless)

Data Unit

- ▶ *Bits*

Summary Table of TCP/IP Layers

Layer	Main Function	Data Unit
Application	User/application communication	Message
Transport	End-to-end process communication	Segment/User Datagram
Network	Routing and host-to-host delivery	Datagram
Data-Link	Link-level delivery	Frame
Physical	Transmission of bits/signals	Bits

TCP/IP Five-Layers

Layer	Function	Responsibilities	Protocols / Devices	Examples
Application Layer	Provides network services directly to user applications.	User communication, data formatting, network services, resource sharing.	HTTP, HTTPS, FTP, SMTP, DNS	Web browsing, Email, File transfer, Domain name lookup
Transport Layer	Provides end-to-end process-to-process communication between hosts.	Segmentation, error control, flow control, reliable delivery, multiplexing.	TCP, UDP	File transfer (TCP), Web browsing (TCP), Video streaming (UDP), Online gaming (UDP)
Network Layer	Handles routing and delivery of packets across networks.	Logical addressing, routing, packet forwarding, path determination.	IP (IPv4/IPv6), Routers	Sending data between different networks, Internet communication
Data Link Layer	Provides communication between devices on the same network link.	Framing, physical addressing (MAC), error detection, media access control.	Ethernet, Wi-Fi, Switches	Communication within a LAN, Frame transmission between devices
Physical Layer	Transmits raw bits through the transmission medium.	Signal transmission, bit representation, cable and connector specifications.	Cables, Hubs, Repeaters, Fiber Optics	Electrical signals through cables, Wireless radio signals, Optical transmission through fiber

Logical Communication

In protocol layering:

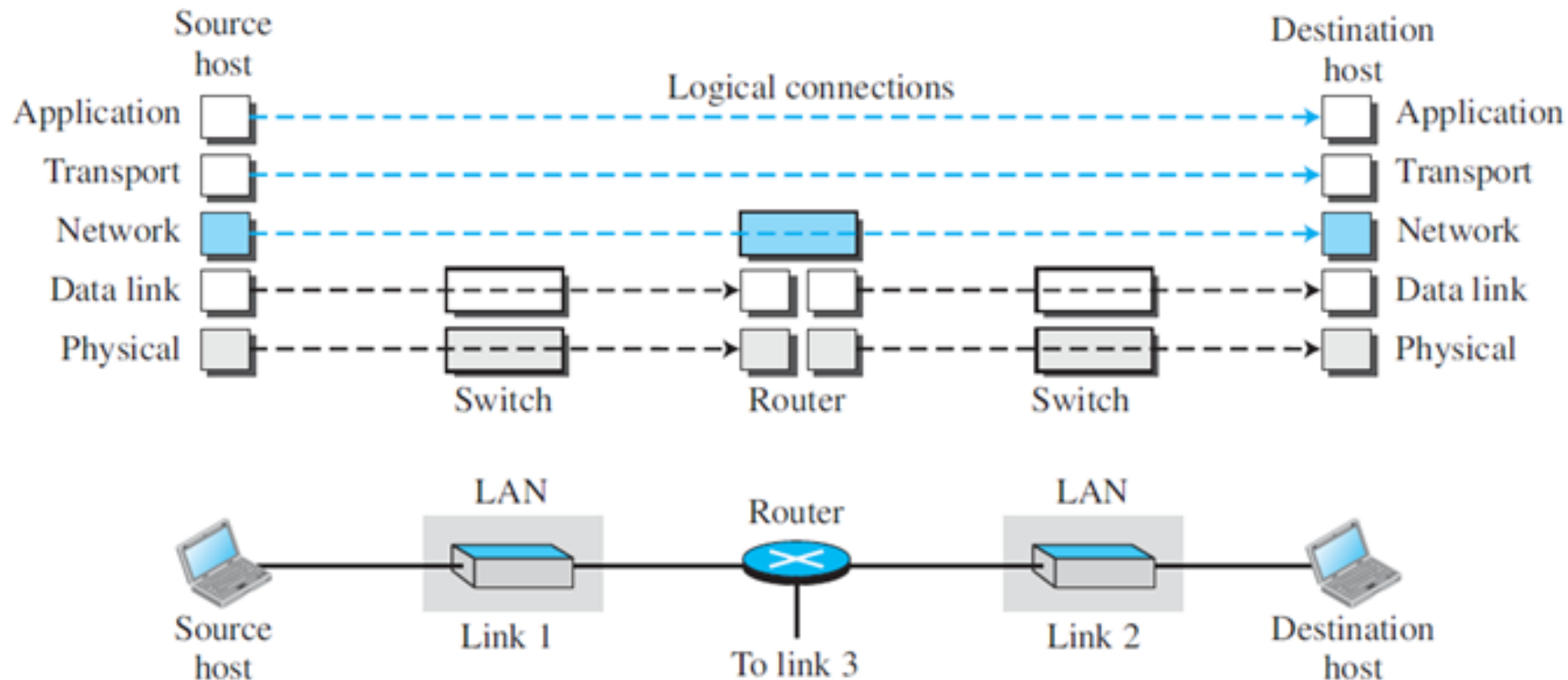
- ▶ Each layer logically communicates with its corresponding layer on the destination host.

Example:

- ▶ Application layer communicates logically with application layer
- ▶ Transport layer co
- ▶ ***Logical Layer-to-Layer Communication***

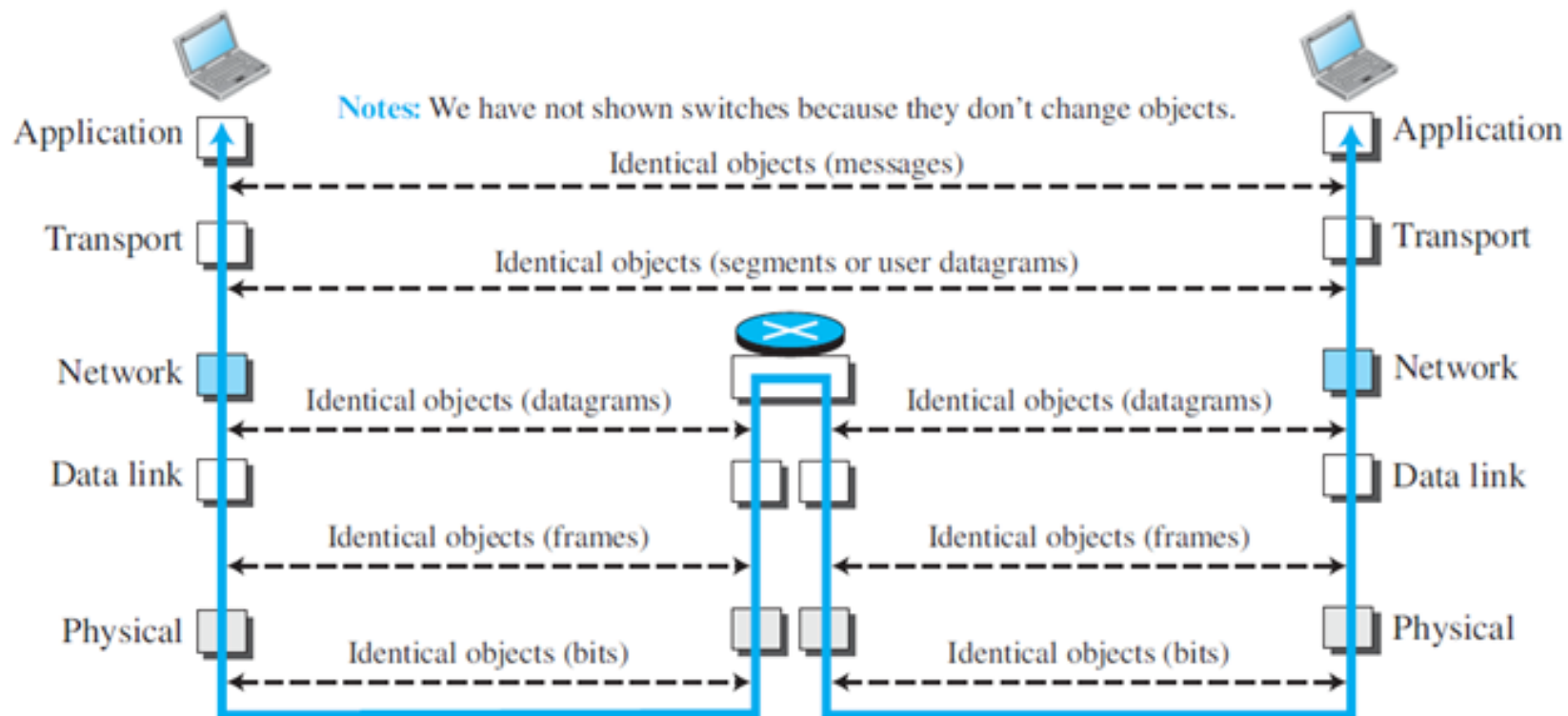
Logical Communication

Figure 1.14 *Logical connections between layers of the TCP/IP protocol suite*



Identical Objects in the TCP/IP Protocol Suite

Figure 1.15 *Identical objects in the TCP/IP protocol suite*



Identical Objects in the TCP/IP Protocol Suite

- ▶ Although data physically travels through different layers and devices, each layer on the sender appears to communicate directly with its corresponding layer on the receiver.

Identical Objects in the TCP/IP Protocol Suite

Application Layer

- ▶ The sender creates a **message**.
- ▶ Logically, the Application Layer communicates with the Application Layer of the destination host.
- ▶ The object exchanged is called a **Message**.

Identical Objects in the TCP/IP Protocol Suite

Transport Layer

- ▶ The message is divided into **segments** (TCP) or **user datagrams**.
- ▶ Logical communication occurs between the Transport Layers of the source and destination hosts.
- ▶ The object exchanged is a **Segment/User Datagram**.

Identical Objects in the TCP/IP Protocol Suite

Network Layer

- ▶ The Transport Layer passes data to the Network Layer.
- ▶ The Network Layer adds an IP header, creating a **Datagram (Packet)**.
- ▶ Routers process data only up to this layer.
- ▶ The datagram remains logically the same throughout the Internet.

Identical Objects in the TCP/IP Protocol Suite

Data Link Layer

- ▶ The Network Layer passes the datagram to the Data Link Layer.
- ▶ The Data Link Layer encapsulates it into a **Frame**.
- ▶ Frames are **exchanged only between directly connected devices**.
- ▶ At every hop, a new frame may be created.

Identical Objects in the TCP/IP Protocol Suite

Physical Layer

- ▶ Frames are converted into **Bits**.
- ▶ Bits are transmitted as electrical, optical, or wireless signals through the medium.

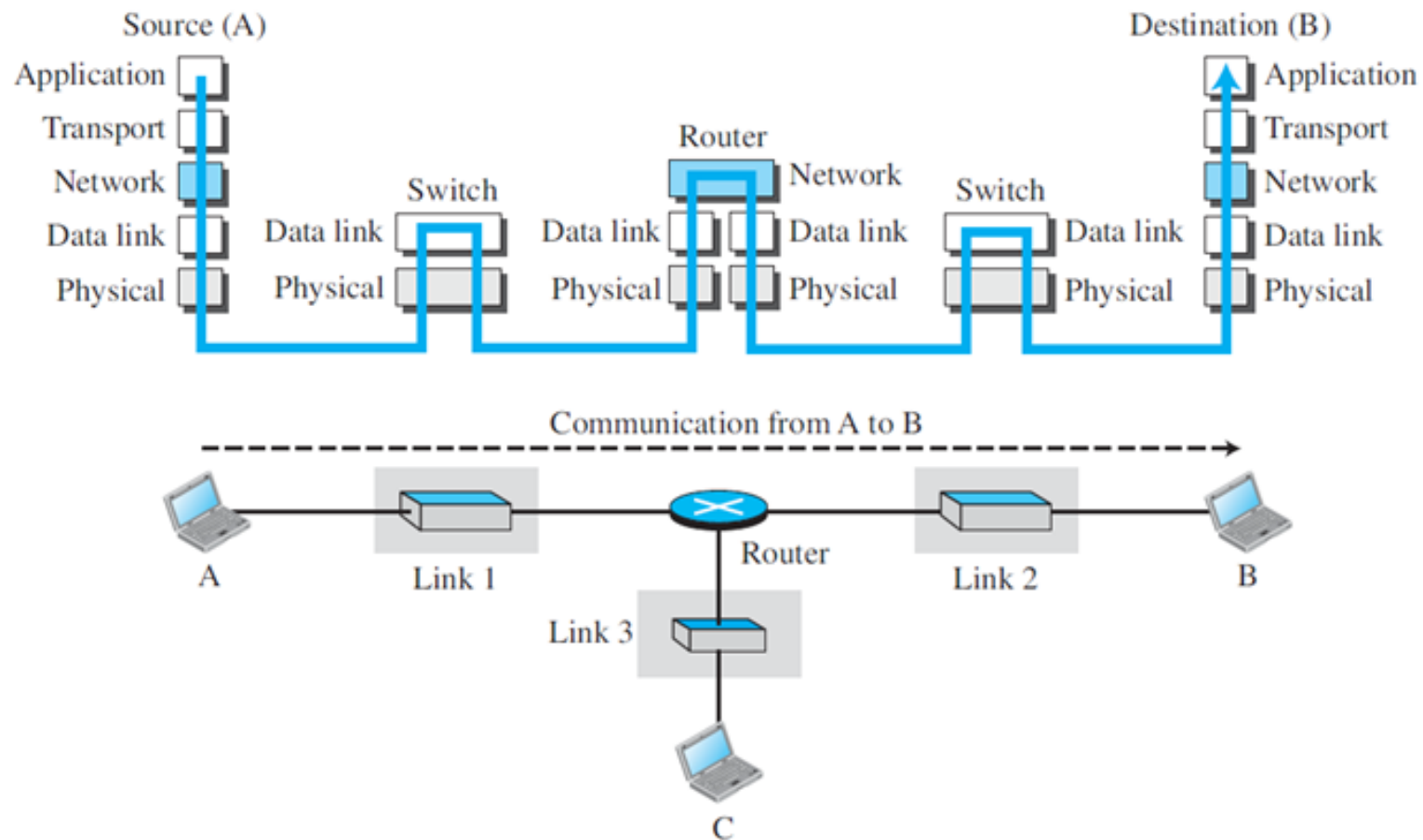
Communication in TCP/IP

Suppose:

- ▶ Computer A sends data to Computer B through switches and a router.
- ▶ The communication passes through all five layers at the sender and receiver.

Communication in TCP/IP

Figure 1.13 *Communication through an internet*



Roles of Devices in TCP/IP

Devices	Layers Used	Function
Host Computer	Application, Transport, Network, Data Link, Physical	Generates and receives data
Router	Network, Data Link, Physical	Routes packets between networks
Switch	Data Link, Physical	Forwards frame within LAN

End-to-End and Hop-to-Hop Communication

End-to-End Communication

- ▶ The top three layers perform: End-to-end communication

These layers are:

- ▶ Application
- ▶ Transport
- ▶ Network
- ▶ They work across the entire Internet.

End-to-End and Hop-to-Hop Communication

Hop-to-Hop Communication

- ▶ The lower two layers perform: Hop-to-hop communication

These layers are:

- ▶ Data Link
- ▶ Physical
- ▶ They work only between directly connected devices.

Advantages of TCP/IP Protocol Suite

Modularity

- ▶ Each layer works independently.

Flexibility

- ▶ Protocols can be updated without affecting all layers.

Scalability

- ▶ Supports large global networks.

Interoperability

- ▶ Allows communication between different systems and devices.

Importance of TCP/IP

TCP/IP is the foundation of:

- ▶ The Internet
- ▶ Modern networking
- ▶ Web communication
- ▶ Cloud computing
- ▶ Almost all modern networks use TCP/IP protocols.

Encapsulation and Decapsulation in TCP/IP

Encapsulation at the Source Host

- ▶ The Application Layer creates the data called a Message.
- ▶ The Transport Layer receives the message and adds a Transport Header, creating a Segment (TCP) or User Datagram (UDP).
- ▶ The Network Layer adds an IP Header to the segment, creating a Datagram (Packet).
- ▶ The Data Link Layer adds a Frame Header, creating a Frame.
- ▶ The Physical Layer converts the frame into Bits and transmits them through the communication medium.

Encapsulation Flow

- ▶ **Message → Segment/User Datagram → Datagram → Frame → Bits**

Encapsulation and Decapsulation in TCP/IP

Decapsulation at Destination Host

- ▶ The physical layer receives the bits
- ▶ The data link layer removes the frames header and extracts the datagram
- ▶ The network layer removes the IP header and extracts the segment
- ▶ The transport layer removes the transport header and extracts the message
- ▶ The application layer receives the original message.

Decapsulation Flow

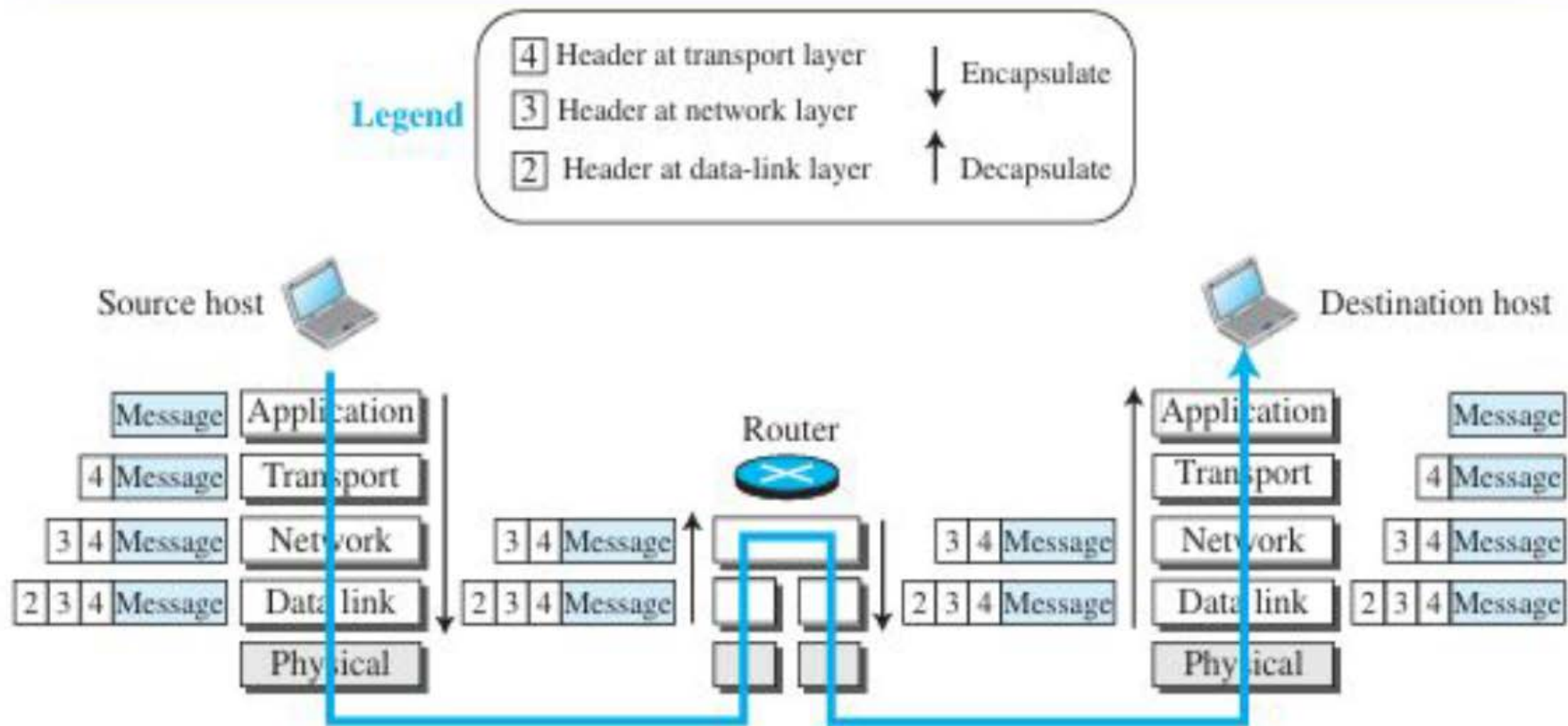
Bits → Frame → Datagram → Segments/User Datagram → Message

Encapsulation and Decapsulation at Routers

- ▶ The router receives the frame and removes the data link header (Decapsulation)
- ▶ The router examines the destination IP address in the datagram
- ▶ The router determines the next hop using its routing table
- ▶ The datagram is then passed to the outgoing data link layer
- ▶ A new frame is created by adding a new data link header (Encapsulation)
- ▶ The frame is transmitted to the next device

Encapsulation and Decapsulation

Figure 1.16 Encapsulation/Decapsulation



Addressing in TCP/IP

- ▶ Communication between two devices requires a source address and a destination address.
- ▶ In the TCP/IP model, different layers use different types of addresses.

Types of Addresses in TCP/IP

Layer	Address Type	Purpose	Example
Application Layer	Names	Identify user or services	www.google.com
Transport Layer	Port Number	Identify application processes	Port 80(HTTP), Port 25 (SMTP)
Network Layer	Logical Address (IP)	Identify hosts on the internet	192.168.1.10
Data Link Layer	Link Layer (MAC) Addresses	Identify devices on a local network	00:1A:2B:3C:4D:5E
Physical Layer	No address	Transmits bits	

Multiplexing and Demultiplexing

Multiplexing

- ▶ Multiplexing occurs at the **source (sender)**.
- ▶ A protocol at a layer can receive data from multiple higher-layer protocols.
- ▶ The protocol encapsulates the data and forwards it to the lower layer.
- ▶ A header field is used to identify the protocol from which the data originated.

Multiplexing and Demultiplexing

Examples

▶ **Transport Layer**

- ▶ TCP can receive data from HTTP, FTP, SMTP, DNS, etc.
- ▶ UDP can receive data from DNS, DHCP, VoIP, etc.

▶ **Network Layer**

▶ IP can receive:

- ▶ TCP Segments
- ▶ UDP User Datagrams
- ▶ ICMP Packets
- ▶ IGMP Packets

Multiplexing and Demultiplexing

- ▶ **Data Link Layer**
- ▶ A frame can carry:
 - ▶ IP Datagrams
 - ▶ ARP Packets
 - ▶ Other network-layer protocols

Demultiplexing

- ▶ Demultiplexing occurs at the **destination (receiver)**.
- ▶ A protocol examines the header information of the received packet.
- ▶ Based on the protocol identifier, the packet is delivered to the correct higher-layer protocol.
- ▶ The packet is then decapsulated and passed upward.

Demultiplexing

Examples

- ▶ **Data Link Layer**

- ▶ Delivers payload to IP or ARP.

- ▶ **Network Layer**

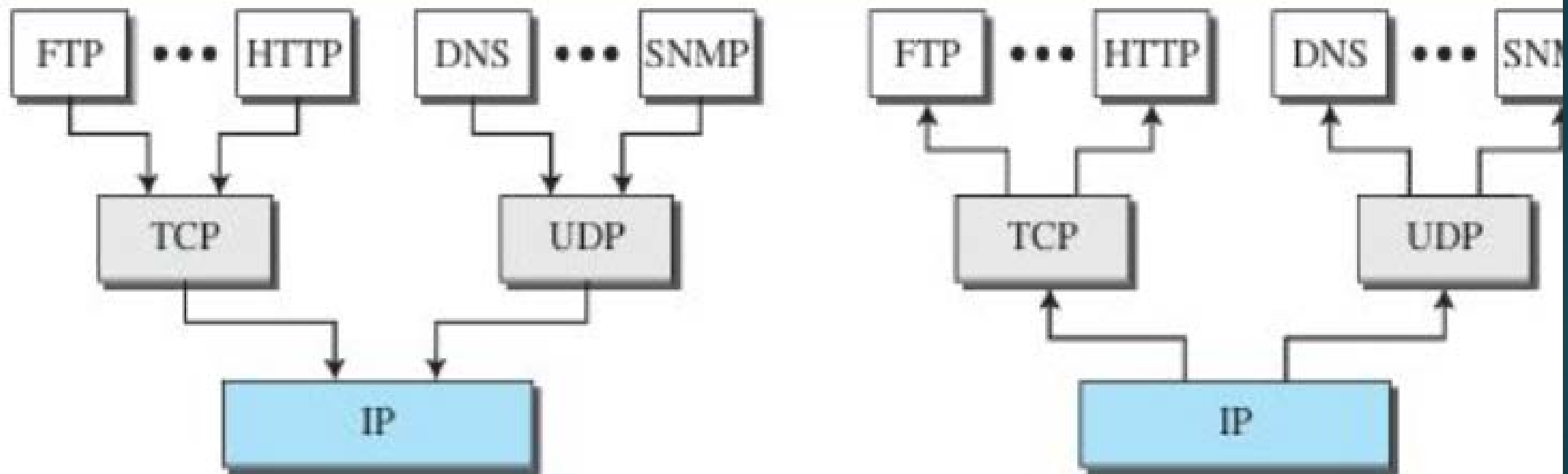
- ▶ Delivers packets to TCP, UDP, ICMP, or IGMP.

- ▶ **Transport Layer**

- ▶ Delivers data to the correct application such as HTTP, FTP, SMTP, or DNS using port numbers.

Multiplexing and Demultiplexing

Figure 1.18 *Multiplexing and demultiplexing*



a. Multiplexing at source

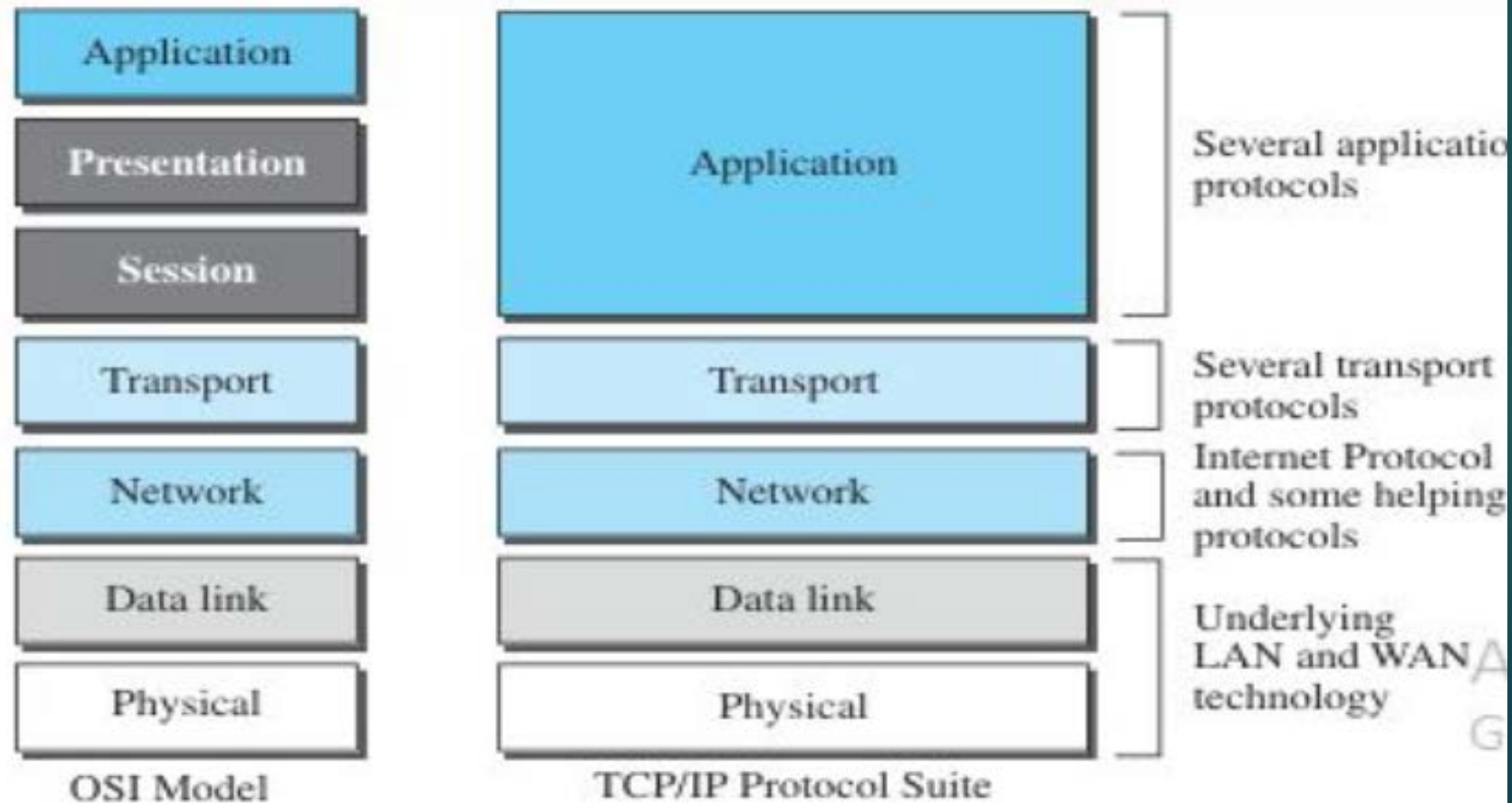
b. Demultiplexing at destination

OSI Model

- ▶ The OSI (Open Source Interconnection) model is a reference model, not a protocol.
- ▶ It provides a framework for designing and understanding network communication
- ▶ The main goal of the OSI model is to enable communication between different computer systems regardless of their hardware and software architecture.
- ▶ The OSI model uses a layered architecture with 7 layers

OSI Model

Figure 1.20 *TCP/IP and OSI model*



OSI Model

1. Application Layer (Layer 7)

- ▶ Provides services directly to users and applications.
- ▶ Examples: HTTP, FTP, SMTP, DNS

OSI Model

2. Presentation Layer (Layer 6)

- ▶ Handles data translation, encryption, and compression.
- ▶ Ensures data is presented in a usable format.

OSI Model

3. Session Layer (Layer 5)

- ▶ Establishes, manages, and terminates communication sessions between applications.

OSI Model

4. Transport Layer (Layer 4)

- ▶ Provides end-to-end communication and reliability.
- ▶ Examples: TCP, UDP

OSI Model

5. Network Layer (Layer 3)

- ▶ Handles logical addressing and routing.
- ▶ Example: IP

OSI Model

6. Data Link Layer (Layer 2)

- ▶ Provides node-to-node communication.
- ▶ Performs framing and error detection.

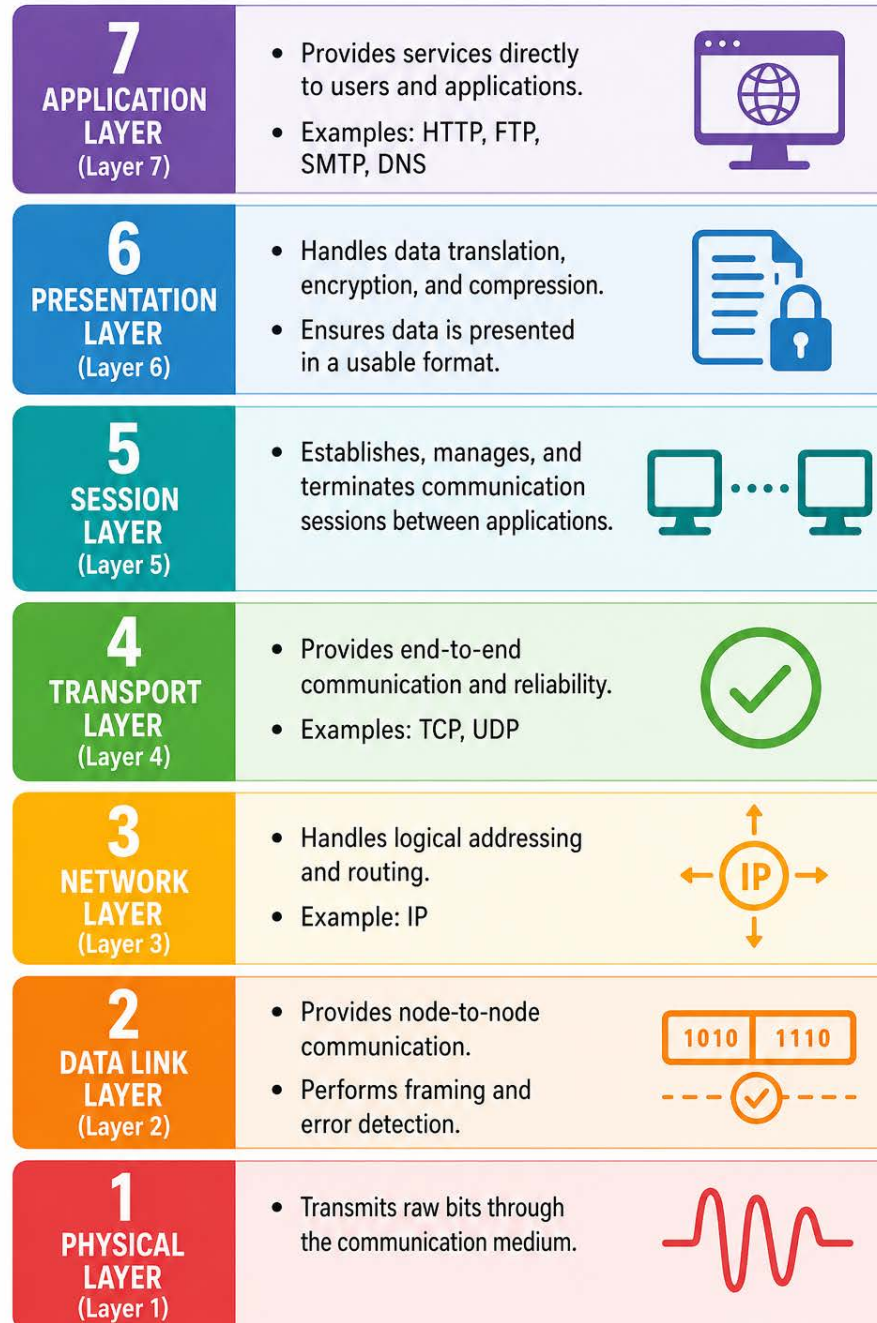
OSI Model

7. Physical Layer (Layer 1)

- ▶ Transmits raw bits through the communication medium.

OSI Model

THE 7 LAYERS OF THE OSI MODEL



OSI vs TCP/IP

Similarities

- ▶ Both are layered models.
- ▶ Both support network communication.
- ▶ Both divide communication into separate functions.

OSI vs TCP/IP

Differences

OSI Model	TCP/IP Model
7 Layers	5 Layers
Reference model	Protocol suite
Developed by ISO	Developed by DARPA/DoD
Presentation and Session layers are separate	Presentation and Session functions are included in the Application layer
Mainly used for learning and design	Used in the Internet and real-world networks

Why TCP/IP Does Not Have Session and Presentation Layers

- ▶ Some Session Layer functions are handled by Transport Layer protocols.
- ▶ Application programs can implement Presentation and Session functions if needed.
- ▶ Combining these layers simplifies the TCP/IP architecture.